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Pisarna 312

1. Mat. indukcija, $n \in \mathbb{N}$

a)

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

1. $n=1$

L: 1

$$D: \frac{1 \cdot 2 \cdot 3}{6} = 1 \checkmark$$

2. $n \rightarrow n+1$

$$\begin{aligned} L: 1^2 + 2^2 + \dots + n^2 + (n+1)^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \\ &= (n+1) \left(\frac{n(2n+1)}{6} + (n+1) \right) = (n+1) \left(\frac{2n^2 + n + 6n + 6}{6} \right) = \\ &= (n+1) \left(\frac{2n^2 + 7n + 6}{6} \right) = (n+1) \frac{(2n+3)(n+2)}{6} \end{aligned}$$

$$D: \frac{(n+1)(n+2)(2(n+1)+1)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}$$

$$L = D \checkmark$$

b)

$$\varphi \in (0, 2\pi)$$

$$1 + \cos \varphi + \cos 2\varphi + \dots + \cos n\varphi = \frac{1}{2} + \frac{\sin((n+\frac{1}{2})\varphi)}{2 \sin \frac{\varphi}{2}}$$

1. $n=1$

$$1 + \cos \varphi = \frac{1}{2} + \frac{\sin \frac{3}{2}\varphi}{2 \sin \frac{\varphi}{2}} = \frac{\sin \frac{\varphi}{2} + \sin \frac{3}{2}\varphi}{2 \sin \frac{\varphi}{2}}$$

$$1 + \cos \varphi = \frac{2 \sin \varphi \cos \frac{\varphi}{2}}{2 \sin \frac{\varphi}{2}} = \frac{2 \cos \frac{\varphi}{2} \cancel{\sin \frac{\varphi}{2}} \cos \frac{\varphi}{2}}{\cancel{\sin \frac{\varphi}{2}}} = 2 \cos^2 \frac{\varphi}{2}$$

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}$$

$$\sin 2\varphi = 2 \cos \varphi \sin \varphi$$

$$\cos 2\varphi = \cos^2 \varphi - \sin^2 \varphi = 2 \cos^2 \varphi - 1$$

$$2 \cos \frac{\alpha}{2} = \cos \frac{\alpha}{2}$$

$$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

(2) I.P. $n \rightarrow n+1$

$$\frac{1}{2} + \frac{\sin((n+\frac{3}{2})\varphi)}{2 \sin \frac{1}{2} \varphi} + \cos((n+1)\varphi) = \frac{1}{2} + \frac{[\sin((n+\frac{1}{2})\varphi) + 2 \cos((n+1)\varphi) \sin \frac{1}{2} \varphi]}{2 \sin \frac{1}{2} \varphi}$$

$$[\sin((n+\frac{3}{2})\varphi)] = [\sin((n+\frac{1}{2})\varphi) + 2 \cos((n+1)\varphi) \sin \frac{1}{2} \varphi]$$

$$\sin \underbrace{(n+\frac{3}{2})\varphi}_{\alpha} - \sin \underbrace{(n+\frac{1}{2})\varphi}_{\beta} = 2 \cos((n+1)\varphi) \sin \frac{1}{2} \varphi$$

$$2 \sin \frac{1\varphi}{2} \cos((n\varphi + 1\varphi))$$

L=D
✓

c) $n! \leq \left(\frac{n+1}{2}\right)^n$

1. $n=1$

$$1 \leq \left(\frac{1+1}{2}\right)^1 = 1 \checkmark$$

(2) I.P. $n \rightarrow n+1$

$$L: \left(\frac{n+1}{2}\right)^n (n+1) = \frac{(n+1)^{n+1}}{2^n}$$

$$D: \left(\frac{n+2}{2}\right)^{n+1} = \frac{(n+2)^{n+1}}{2^{n+1}} = \frac{(n+2)^{n+1}}{2 \cdot 2^n}$$

$$\frac{(n+1)^{n+1}}{2^{n+1}} \leq \frac{(n+2)^{n+1}}{2 \cdot 2^n}$$

$$2 \cdot (n+1)^{n+1} \leq (n+2)^{n+1}$$

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

$$\frac{(n+1)+1}{n+1}$$

$$\uparrow$$

$$n+1$$

$$2 \leq \frac{\binom{n+2}{1}}{\binom{n+1}{1}} = \left(\frac{n+2}{n+1} \right) = \left(1 + \frac{1}{n+1} \right)$$

$$= 1 + \underbrace{\binom{n+1}{1} \frac{1}{n+1}}_1 + \dots$$

$$\binom{n+1}{1} = \frac{(n+1)!}{n! 1!} = n+1$$

② Bijekcija (poišči zvezo, da bo f bijektivna)

a) $f: \mathbb{N} \rightarrow \mathbb{N} \cup \{0\}$

$$f(n) = n - 1$$

$$g(n) = n + 1$$

b) $f: \mathbb{N} \rightarrow 5\mathbb{N}$

$$f: n \mapsto 5n$$

$$g: n \mapsto \frac{1}{5}n$$

$$g: 5\mathbb{N} \mapsto \mathbb{N}$$

c) $f: \mathbb{N} \rightarrow \mathbb{Z}$

$$1 \mapsto 0$$

$$2 \mapsto 1$$

$$3 \mapsto -1$$

$$4 \mapsto 2$$

$$5 \mapsto -2$$

⋮

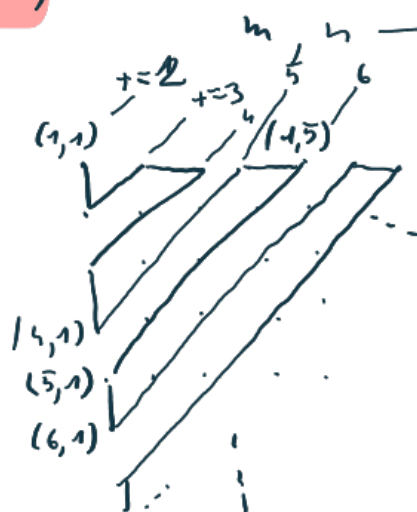
$$f(n) = \begin{cases} \frac{n}{2} & ; n \text{ sodo} \\ -\frac{n-1}{2} & ; n \text{ liho} \end{cases}$$

$$g(n) = \begin{cases} 2n & ; n > 0 \\ -2(n)+1 & ; n \leq 0 \end{cases}$$

$$g: \mathbb{Z} \rightarrow \mathbb{N}$$

③. Bijekcija

a) $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$



$$1, 1 \rightarrow 1$$

$$2, 1 \rightarrow 2$$

$$1, 2 \rightarrow 3$$

$$1, 3 \rightarrow 4$$

$$2, 2 \rightarrow 5$$

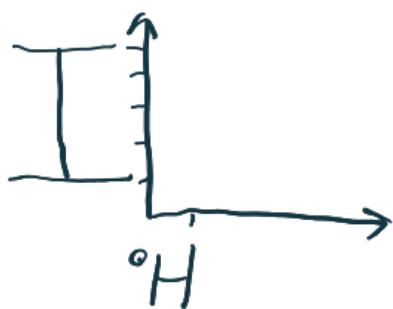
$$3, 1 \rightarrow 6$$

$$f(m, n) = 1 + 2 + \dots + (m+n-2) + m$$

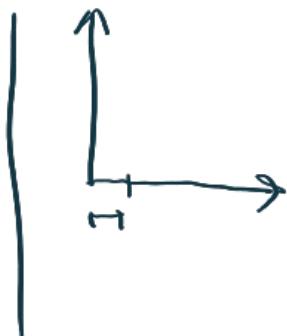
4. Bijekcija

a) $f: (0, 1) \mapsto (1, 5)$ $f(x) = 4x + 1$

$$f^{-1}(x) = \frac{1}{4}(x-1)$$



b) $f: (0, 1) \rightarrow \mathbb{R}$



$$f(x) = \tan\left(\pi\left(x - \frac{1}{2}\right)\right)$$

$$x = \tan\left(\pi\left(y - \frac{1}{2}\right)\right)$$

$$\pi\left(y - \frac{1}{2}\right) = \arctan x$$

$$f^{-1}(x) = \frac{1}{\pi} \arctan x + \frac{1}{2}$$

c) $f: (-5, \infty) \rightarrow \mathbb{R}$

↑

$$f(x) = \ln(x+5)$$



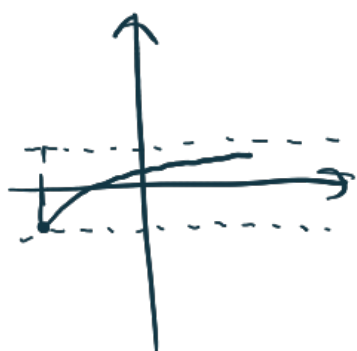
$$x = \ln(y+5)$$

$$e^x = y+5$$

$$\underline{f^{-1}(x) = e^x - 5}$$

d)

$$f: [-5, \infty) \rightarrow [-1, 1] \quad \text{rac.}$$



$$f(x) = -\frac{2}{x+6} + 1$$

5.

Bijekcija

a)

$$f: [0, 1) \rightarrow (0, 1)$$

$$f(x) = \begin{cases} x; & x \notin A \\ g(x); & x \in A \end{cases}$$

! f ni zvezha!

$$A = \left\{0, \frac{1}{n}; n \in \mathbb{N} \setminus \{1\}\right\}$$

$$A' = \left\{\frac{1}{n}; n \in \mathbb{N} \setminus \{1\}\right\}$$

$$g: A \rightarrow A'$$

$$g(x) = \begin{cases} \frac{1}{2}; & x = 0 \\ (x^{-1}+1)^{-1}; & x \neq 0 \end{cases}$$

b) $f: [0, 1] \rightarrow (0, 1)$

$$A = \left\{0, \frac{1}{n}; n \in \mathbb{N}\right\}$$

$$f(x) = \begin{cases} g(x); & x \in A \\ x; & x \notin A \end{cases}$$

$$g: A \rightarrow A'$$

$$g(x) = \begin{cases} \frac{1}{2}; & x = 0 \\ \frac{1}{3}; & x = 1 \\ (x^{-1}+2)^{-1}; & x \neq 0, x \neq 1 \end{cases}$$

6. Bijekcija

$$f: \underset{a \in \mathbb{Q}}{[0,1)} \times \underset{b \in \mathbb{Q}}{[0,1)} \rightarrow [0,1)$$

$$\left. \begin{array}{l} a = 0, a_1 a_2 a_3 \dots \\ b = 0, b_1 b_2 b_3 \dots \end{array} \right\} \text{ gde se ne pojavljuje } \infty \text{ u nesk.}$$

$$f(a, b) = 0, a_1 b_1 a_2 b_2 a_3 b_3 \dots$$

Ali je bijekcija?

Cantor Bernstein-Schroederjev izrek:

če je $A \rightarrow B$ inj. in

$B \rightarrow A$ inj.;

potem je $A \rightarrow B$ bijekt.

1. inj. $f(x)$
2. inj. $h(x) = (x, a)$
 $\Downarrow$
je bijekt.

7. Pokaži, da sta ekvipotentni:

a) $(0,1) \times (0,1)$; $(0,1)$

$$\left(\begin{array}{l} - |A| = |B| \wedge |B| = |C| \Rightarrow |A| = |C| \\ - |A| = |A'| \wedge |B| = |B'| \Rightarrow |A \times B| = |A' \times B'| \end{array} \right)$$

- Prejšnje naloge:

$$- |[0,1) \times [0,1)| = |[0,1)|$$

$$- |[0,1)| = |(0,1)|$$

$$- |\mathbb{R}| = |(0,1)|$$



$$| \overset{A}{(0,1)} \times \overset{B}{(0,1)} | = | \overset{A'}{[0,1)} \times \overset{B'}{[0,1)} | = |[0,1)| = |(0,1)| \quad \checkmark$$

b) $\mathbb{R}$; $\mathbb{R}^n$; $n \in \mathbb{N}$

① $n=1$ $|\mathbb{R}| = |\mathbb{R}^1| \quad \checkmark$

② $n \rightarrow n+1$ I.P.

$$|\mathbb{R}^{n+1}| = |\mathbb{R}^n \times \mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |(0,1) \times (0,1)| =$$

$\uparrow$
2. av. I.P.

$$= |(0,1)| = |\mathbb{R}| \quad \checkmark$$

8. Dokaži:
 $f: X \rightarrow Y$ $g: Y \rightarrow Z$

a) f inj. $\wedge$ g inj. $\Rightarrow g \circ f$ inj.

vzemimo $x_1, x_2 \in X$; $x_1 \neq x_2$

$$\text{Ker } f \text{ inj.} \Rightarrow f(x_1) \neq f(x_2) \xRightarrow{g \text{ inj.}} g(f(x_1)) \neq g(f(x_2)) \Rightarrow \\ \Rightarrow g \circ f \text{ je inj}$$

b) $f \text{ surj.} \wedge g \text{ surj.} \Rightarrow g \circ f$

vzemimo $\forall z \in Z; \exists y \in Y: \underline{g(y) = z};$

$\exists x \in X: \underline{f(x) = y} \Rightarrow g(f(x)) = z \Rightarrow \\ \Rightarrow g \circ f \text{ je surj.}$

c) $g \circ f$

